

Intrusion Detection in IoT leveraged by Multi-Access Edge Computing using Machine Learning

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Abstract— An intrusion detection system is a technology built to monitor, detect, and prevent malicious activities and cyberattacks towards computer networks. The Internet of Things (IoT) is a system of devices connected to each other and to the Internet that facilitates communication between the IoT devices and the cloud. This rapidly growing technology calls for more accurate and efficient cyberattack detection techniques to ensure the security of IoT devices and systems. In this paper, we focus on using machine learning and deep learning algorithms along with feature selection methods to detect cyberattacks effectively at the edge of IoT network by leveraging multi-access edge computing. This study addresses the challenges of processing intrusion detection datasets, such as data imbalance and missing data, with a focus on UNSW-NB15 dataset. We use ANOVA and embedded feature selection techniques and apply various machine learning algorithms, consisting of Decision Tree (DT), Random Forest (RF), Light gradient-boosting machine (LightGBM), Artificial Neural Network (ANN), K-Nearest Neighbor (kNN), and Extreme Gradient Boost (XGB), on UNSW-NB15 dataset. Our experimental results indicate that our classifiers achieved higher accuracies and efficiencies comparing to state-of-the-art machine learning intrusion detection approaches and our LightGBM model is the most accurate and efficient one among all.

Keywords— *Intrusion detection; Multi-Access Edge Computing; Network security; Feature Selection; Machine Learning;*

I. INTRODUCTION

With the excessive number of IoT devices in the modern life, the potential impact of cyberattacks has become significant. Such attacks can disrupt companies and individuals' work and impose substantial financial burdens on businesses and society as a whole. To safeguard devices and organizations against these threats, the development of intrusion detection systems with high accuracy and performance is crucial.

The IoT systems have unique attributes comparing to traditional Internet technology, mainly there is minimum or no human interaction with IoT systems. Also, excessive number of IoT devices results in high demands of bandwidth. The CISCO research team predicts almost 75.3 billion active IoT devices will be connected by 2025 [1]. On the other hand, the resource constraints on most IoT devices are a challenge in employing advanced security protection tools. To overcome these challenges, the multi-access edge computing (MEC) platform has been proposed to relocate the complex security protection tasks from the device to the network edge [2]. An alternative approach is to deploy an accurate but less complex system on the IoT device that detects the intrusion on the IoT device and prevents the distribution of malicious code to the rest of IoT network [3]. In this paper, we focus on developing an MEC platform using accurate and efficient machine learning models.

A Network Intrusion Detection System (NIDS) monitors the network traffic, identifies risks, and protects the network from malicious attacks. Usually, the cyber-attacker initially compromises a vulnerable IoT device in the network and then distributes the malicious code to the rest of IoT network and/or the cloud [4]. Detecting the malicious traffic circulating in the network and throttling the attack at that first step at the network edge saves the entire network.

The Mobile Edge Computing (MEC) platform, aka multi or mobile access edge computing [5], has emerged as a solution for IoT device's resource constraints by relocating the computationally complex tasks from the IoT devices to the network edge. The architecture of an MEC platform consists of three layers.

1. IoT devices (e.g., wearable devices, smart thermostat, baby monitoring camera) and Mobile devices (e.g., smart phones, smart watches, tablets, laptops): This is the bottom layer of the MEC platform architecture. The devices are grouped in an IoT network based on their location, ownership, use cases, etc.
2. MEC/Fog: MEC transmits and processes data between IoT devices and the cloud in near real time. MEC system directly connects to the IoT gateway. It is equipped with high performance computing. It oversees computationally complex tasks (e.g., NIDS for the devices in the connected IoT network.)
3. Cloud: The cloud [6] is the network core, consisting of cloud servers and data centers that can process, transmit, and store massive volumes of data.

The current challenges of IoT security are rooted in similar historical security issues of computers and computer networks in mid-1990s when the cyberattacks and threats started to grow against the vulnerabilities of personal computers and their software, and there was no direct approach to fix these security issues. Nowadays, the scale of using IoT is rapidly growing and there is a variety of IoT devices and IoT software with different use cases. The standards of manufacturing, software development, and update management are not uniformly defined, accepted, or applied by the IoT community or manufacturers. That is the main security challenge of IoT.

Different security solutions are adopted based on the use cases and system requirements. In many cases, the IoT device is not the target of the cyberattack, but it is infected to be used in distributed attacks that need massive sources of traffic. The DDoS attack that disrupted Internet in 2016 [7] was the largest of this kind.

In such cases, since the malware infection does not disrupt the regular tasks of the IoT device, the manufacturer or the user may not consider or even be aware of disruptive impacts of their unsecure devices on the Internet. In recent years, especially during the Covid-19 pandemic, there has been a rise of cyberattacks. A variety of advanced attacks in large scale has occurred, which resulted in financial and privacy harms for industry and individuals. The main security challenge in the IoT community is that most IoT manufacturers avoid adopting advanced security solutions to keep the manufacturing costs low. Also, most IoT users ignore installing or updating security protection software on their devices due to the reasons mentioned above. The long-term solution is to define uniform security standards world-wide for all IoT manufacturers.

However, at this point that new versions and more advanced cyberattacks emerge very often, the IoT technologies are developing fast, and the Internet traffic is enormously growing, there is a need for a prompt action to create a robust solution for IoT security issues. MEC platform is that solution [8]. With MEC, disregarding the IoT architecture or software, the computational resources at the edge of the IoT network can secure the network and block malware distribution into the network. In case IoT devices are already infected by malware, the MEC platform can detect the malicious traffic generated from its network and stop it or mitigate its impacts. In our research, we propose machine learning models to leverage intrusion detection at the MEC layer.

The main contributions of our paper are as following.

1. We used the UNSW-NB15 dataset [9]. This dataset is imbalanced. 87.35% of records are benign and the rest are attacks. **To address the data imbalance issue**, we applied Random Under Sampling, experimenting with various post-sampling label distribution ratios. This allowed us to compare the performance of different strategies and select the most effective one for dealing with the dataset's imbalance. Our problem is using binary classification to detect malicious traffic.
2. We applied filter and embedded **feature selection techniques** to obtain features with the highest impact on accurate classification. We employed a comprehensive set of six classification algorithms, namely DT, RF, LightGBM, kNN, ANN, and XGBoost, and thoroughly evaluated their performances on the dataset. Our experimental results indicated that our models outperform the state of the art.

The rest of this paper is organized as the following. Section 2 discusses related intrusion detection work on UNSW-NB15 dataset. Section 3 explains our methodology, including the dataset and our data preprocessing. Section 4 describes the feature engineering methods that we used. Section 5 describes the performance measures. Section 6 explains our strategy to deal with data imbalance in the dataset. Section 7 discusses the results. Section 8 draws the conclusion.

II. RELATED WORK

In our study, we focus on evaluating the performance of an intrusion detection system using the UNSW-NB15 dataset.

Zhou, et al. [10] proposed a framework to detect cyberattacks vs. benign using Deep Feature Embedded Learning (DFEL). They evaluated their framework on two datasets: NSL-KDD and UNSW-NB15. They created a variety of models, including Gradient Boost Tree (GBT), kNN, DT, Logistic Regression (LR), Gaussian Naïve Bayes (GNB), and SVM on UNSW-NB15. They achieved 93% accuracy on GBT using DFEL.

Jing, et al. [11] proposed a new scaling method for intrusion detection based on SVM. In their study, they implemented a nonlinear log function scaling method rather than linear normalization. They demonstrated that a nonlinear log function scaling method could overcome the limitations in the traditional linear normalization and is more suitable for the UNSW-NB15 dataset. They achieved an accuracy of 85.99% for binary classification and 83.59% for multi class classification.

Kasongo, et al. [12] presented a performance analysis of both binary and multi class classification on the UNSW-NB15 dataset. They applied a filter based feature selection technique using XGBoost algorithm. By comparing performances of SVM, kNN, LR, ANN, and DT, with and without feature selection, they observed that feature selection improves the accuracy. For instance, the accuracy of their DT improved from 88.13% to 90.85% with feature selection using XGBoost.

Kabir, et al. [13] proposed stacking machine learning models instead of individual models for the intrusion detection system on UNSW-NB15. They explored various stacking combinations for machine learning models and different feature selection methods, and found that the stacking model built with XGBoost, kNN, and RF gives better accuracy with Mutual Information Gain FS. They found that the stacking model built with XGBoost, kNN, and Random Forest (RF) achieved better accuracy using the Mutual Information Gain FS feature selection method. This combination resulted in an accuracy of 96.24%, outperforming individual models like XGBoost, RF, kNN, and NN.

Ahmad et al. [14] also worked on UNSW-NB15 dataset and made improvements in accuracy. They presented 4 feature clusters for both binary and multi class classification and applied RF, ANN, and SVM models on the data. They achieved 98.6% accuracy using RF in binary classification. In this dataset, 87% of the records are benign traffic and the rest are attack. They reduced the benign records to 50% and merged them to the attack records. With this reduction, they got a more balanced dataset of almost 77% benign and 23% attack traffic. Then they applied RF, SVM, and ANN to this dataset.

In our research, we used a different approach to get a balanced dataset. We randomly sampled the same number of benign records as the number of attack records in the dataset and merged them together. Therefore, we got a perfectly balanced dataset of 50% benign and 50% attack records, which enhanced the robustness of our approach. Our problem is a binary classification to detect attack traffic. Furthermore, we applied feature selection techniques to choose features that have the highest impact on accurate classification. We applied 7 classification algorithms on the dataset, which are RF, SVM, LightGBM, kNN, ANN, DT, and XGBoost.

III. METHODOLOGY

A. Dataset

The UNSW-NB15 dataset consists of 2,540,044 records [9], 87.35% of them are benign records and the remaining 12.65% are attack records. It contains 49 columns, consisting of 47 features, one attack category, and one label (benign or attack). Five features are of nominal data type, which we removed. The remaining features have either integer or float data types. The remaining features have either integer or float data types. Regarding missing values, the columns sport, dsport, ct_flw_http_mthd and ftp_login contain 8,304, 1,348,143 and 1,429,877 missing values, respectively.

B. Data preprocessing

Removing irrelevant columns: First, we removed the five features with nominal data types, srcip (source IP), dstip (destination IP), proto (Protocol), state, and service. These features are also irrelevant to the traffic type. The network traffic has been collected in a test network with fix source and destination IP addresses and protocols, and this information do not present useful data to classify the traffic type. We also removed the attack category column, which is used for multi-class classification.

To handle missing values, [14] used three imputation methods, which are mean, multiple and linear regression. They did not find any significant difference between these three methods. Using their results as a reference, we only used mean imputation to handle missing values in our work.

The UNSW-NB15 dataset is imbalanced. 87.35% of records are benign records and the rest are attack records. Imbalanced datasets can negatively impact the model's performance. We applied Random Under Sampling [10] on the dataset to reduce the number of benign records. Then, we did thorough examination of undersampling on the dataset to find the best strategy to deal with data imbalance.

We first split the dataset to 70% train set and 30% test set. Then, we studied 4 different post sampling class distribution ratios on the training set using Random Under Sampling. We studied 4 post sampling class distribution ratios as following.

1. 50:50 malware: benign (Ratio 50)
2. 33:67 malware: benign (Ratio 33)
3. 23:77 malware: benign (Ratio 23)
4. 13:87 malware: benign (Ratio 13)

The Ratio 50 training set is a perfectly balanced dataset with smallest data size, while the Ratio 13 training set is the original training set without Random Under Sampling. We began with Ratio33 and performed feature selection and model training. We achieved the best performance with 15 features. Then, we studied other ratios with 15 selected features and compared results on three models, which are Random Forest, LightGBM, and XGBoost models. The details are presented in the next sections.

Z-score standardization technique was used to convert each of the variables into a similar scale by centering each variable at zero with a standard deviation of 1.

Table 1: Selected features after applying ANOVA F-test feature selection. Features are sorted based on F-score values.

Feature	F-score
Feature 10: sttl	7,992,620
Feature 37: ct_state_ttl	5,737,922
Feature 47: ct_dst_src_ltm	427,958
Feature 46: ct_dst_sport_ltm	379,076
Feature 45: ct_src_dport_ltm	332,989
Feature 42: ct_srv_dst	313,357
Feature 41: ct_srv_src	306,894
Feature 44: ct_src_ltm	238,113
Feature 43: ct_dst_ltm	232,206
Feature 19: swin	196,571
Feature 20: dwin	194,940
Feature 29: Stime	145,847
Feature 30: Ltime	145,847
Feature 24: dmeansz	142,616
Feature 22: dtcpb	102,782

IV. THE FEATURE SELECTION TECHNIQUES

Feature engineering plays a crucial role in improving the performance of machine learning models by identifying and selecting the most relevant features from a dataset. The authors of [14] created four feature clusters based on the type of features. In our paper, we studied two feature selection methods: (1) filter and (2) embedded.

For filter method, we studied ANOVA (analysis of variance) F-test as a feature selection technique to extract the most important features. When applying the ANOVA F-test, we evaluated different numbers of selected features and determined that 15 features yielded optimal performance. We trained models with less and greater numbers of features, but 15 presented the best performance. The 15 selected features are presented in Table 1.

Feature selection process can be integrated in the model training phase, allowing concurrent execution of model training and feature selection. Tree based models such as Random Forest and XGBoost have their embedded feature selection. For Random Forest, 9 features were selected, which are sbytes, sttl, dttl, Sload, Dload, Dpkts, smeansz, dmeansz, and ct_state_ttl. For XGBoost 4 features were selected, which are dsport, sttl, is_sm_ips_ports, ct_state_ttl.

Table 2: Confusion Matrix

	Predicted Malicious (1)	Predicted Benign (0)
Actually Malicious (1)	TP	FN
Actually Benign (0)	FP	TN

V. PERFORMANCE MEASURES

To analyze the effectiveness of machine learning algorithms, it is important to understand the metrics used to measure the performance of these models.

Confusion matrix: Table 2 presents our confusion matrix.

1. TP = True Positive, malicious traffic was correctly predicted to be malicious.
2. FP = False Positive, benign traffic was incorrectly predicted to be malicious.
3. FN = False Negative, malicious traffic was incorrectly predicted to be benign.
4. TN = True Negative, benign traffic was correctly predicted to be benign.

Accuracy: Accuracy measures the proportion of the correctly predicted emails as spam or ham to the total number of predictions. It is measured by equation 1.

$$Accuracy = \frac{TP+TN}{TP+FP+FN+TN} \quad (1)$$

Precision: Precision measures the proportion of the correctly predicted spam emails to the total number of emails that were predicted to be spam, either correctly or incorrectly. It is measured by equation 2.

$$Precision = \frac{TP}{TP+FP} \quad (2)$$

Recall: Recall measures the proportion of the correctly predicted spam emails to the total number of spam emails that were provided. It is measured by equation 3.

$$Recall = \frac{TP}{TP+FN} \quad (3)$$

F1 score: F1 score measures the ‘Harmonic mean’ of precision and recall values. It is measured by equation 4.

$$F1\ score = \frac{2 \times (precision \times recall)}{precision + recall} \quad (4)$$

VI. DEALING WITH DATA IMBALANCE

We studied 4 post sampling class distribution ratios, consisting of ratio50, ratio33, ratio23, and ratio13. As described in section 3.B, ratio13 means 13% malware and 87% benign records, which is the portions of records in the original dataset. Other ratios are extracted by Random Under Sampling method applied to the dataset to make it more balanced.

We evaluated the performances of the 4 different ratios based on accuracy with RF, LightGBM, and XGBoost models. The results are presented in Table 3. Fig. 1 and 2 plot the accuracy and training time of the three models with different ratios, and Fig. 3 plots the training set size after Random Under Sampling. Although addressing data imbalance usually provides better performance, for this dataset all three models achieved the best accuracy with no undersampling, which is ratio13 on the training set. This could be associated with the training set size. As ratio13 uses the entire dataset and does not remove any data, the training creates the most accurate model. For ratio50, RF and XGBoost presented high numbers of False predictions, resulted in low accuracies of these models.

Table 3: Accuracy comparison of different post sampling ratios

Model	Ratio50	Ratio33	Ratio23	Ratio13
RF	0.324	0.988	0.988	0.992
LightGBM	0.826	0.974	0.988	0.991
XGBoost	0.291	0.986	0.987	0.991

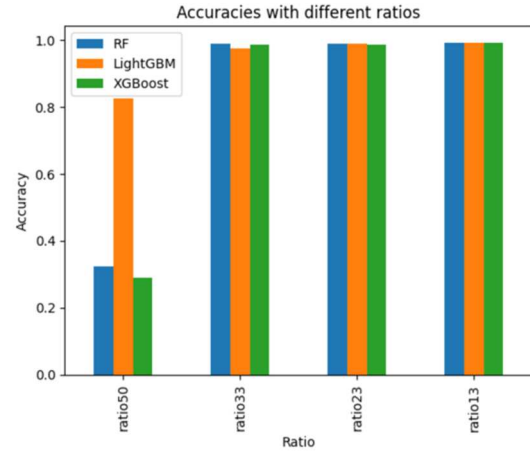


Fig. 1. Accuracy comparison of different ratios

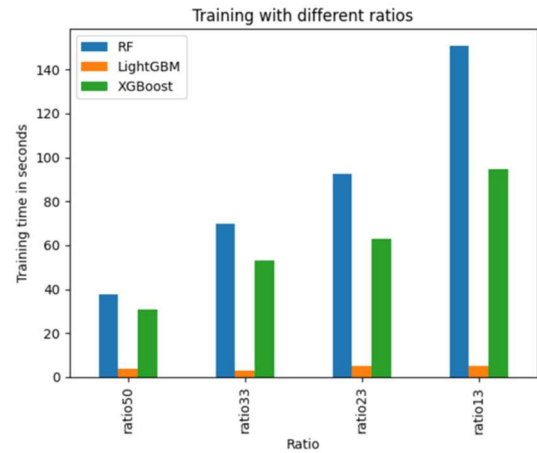


Fig. 2. Training time comparison of different ratios (sec)

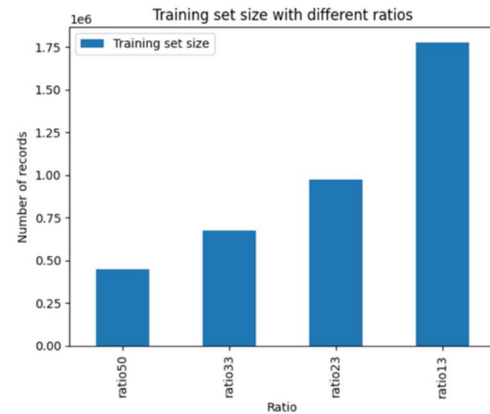


Fig. 3. Training set size comparison of different ratios

Table 4: Binary classification results from previous work on UNSW-NB15 dataset

No	Models	Accuracy [10]	Accuracy [14]	Accuracy [12]	Accuracy [13]	Accuracy [11]
1	RF		0.9867 [14]			
2	SVM	0.9232 [10]	0.9769 [14]	0.6089 [12]		0.8599 [11]
3	ANN		0.9478 [14]	0.8439 [12]		
4	LR	0.9235 [10]		0.7764 [12]		
5	kNN	0.919 [10]		0.8446 [12]		
6	DT	0.9229 [10]		0.9085 [12]		
7	GBT	0.9313 [10]				
8	GNB	0.9252 [10]				
9	Stacking model (built with XGBoost, kNN, and RF)				0.9624 [13]	

Table 5: The proposed models' results

No	Feature selection	Model	Accuracy	Precision	Recall	F-1 score	Training time (sec)	Prediction time (sec)
1	ANOVA	RF	0.992	0.97	0.96	0.97	150.86	6.67
2		LightGBM	0.991	0.97	0.96	0.97	4.99	1.11
3		ANN	0.992	0.97	0.97	0.97	200.04	2.54
4		DT	0.989	0.96	0.95	0.96	5.15	0.09
5		kNN	0.991	0.96	0.97	0.96	7.23	1151.89
6		XGBoost	0.991	0.98	0.95	0.96	94.51	0.83
7	Embedded	RF	0.989	0.99	0.93	0.96	140.33	4.68
8		XGBoost	0.990	0.97	0.95	0.96	56.51	0.58

VII. RESULTS AND DISCUSSION

We chose ratio13, the ratio that yields the best accuracy, as the best ratio for our models. Then, we conducted a comprehensive evaluation of the classification algorithms used in our intrusion detection platform, measuring their accuracy, precision, recall, F1 score, training time, and prediction time. For better comparison with state-of-the-art, some of the results from previous research are summarized in Table 4, while our results are presented in Table 5.

Table 4 indicates that the most accurate binary classification model is RF proposed by [14] with 98.67% accuracy. Table 4 indicates that all our models present higher accuracies than the best model proposed by the previous work. Overall, the accuracies of all our ANOVA models are higher than Embedded models. RF with ANOVA presents higher accuracy (99.2%) than RF with embedded feature selection (98.9%), however, its training and prediction times are higher as well. The same pattern is observed for XGBoost with ANOVA and embedded feature selections.

RF with embedded has relatively low recall (93%). The reason is this model has relatively high False Negative predictions, meaning it incorrectly detects attacks as benign. This results in relatively low accuracy (98.9%) in this model as well.

RF and ANN with ANOVA present the highest same accuracy (99.2%), but RF presents less training time and more Prediction time. Comparing to these two models, LightGBM with ANOVA presents slightly less accuracy (99.1%) but significantly less training time (4.99 sec) and less prediction time (1.11 sec). Comparing to kNN and XGBoost with ANOVA, which have the same accuracy, again LightGBM has significantly less training time. We select LightGBM as the best model for an IoT system. However, selecting one model as the best one depends on the application, and whether the model is being used at the IoT device or at the network edge, thus how much computational resources are available for training and testing.

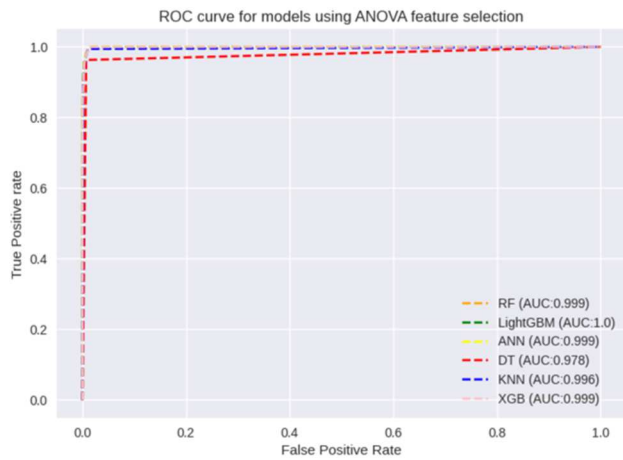


Fig. 4. ROC curves for models that use ANOVA feature selection.

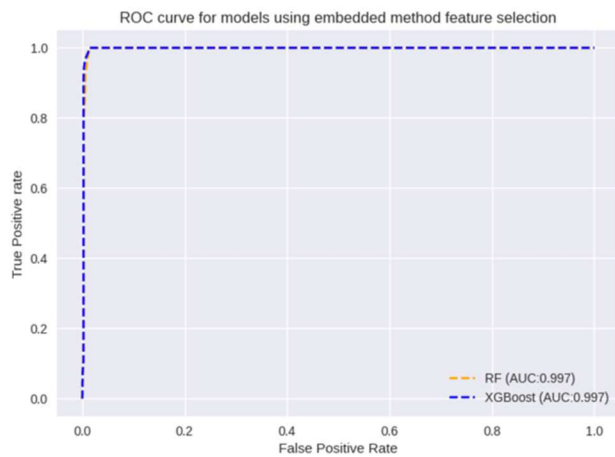


Fig. 5. ROC curves for models that use Embedded feature selection.

An ROC curve plots two parameters, which are true positive rate and false positive rate. AUC refers to the area under the ROC curve, and it is a measure of the quality of models. AUC ranges from 0 to 1, and AUC of 1 means that the model can make the most accurate predictions. Fig. 4 illustrates the ROC curves for RF, LightGBM, ANN, DT, kNN, and XGBoost with ANOVA feature selection. Fig. 5 illustrates the ROC curves for RF and XGBoost using embedded feature selection. Fig. 4 shows that LightGBM with ANOVA feature selection has the highest AUC of 1, another evidence that this is our best model.

VIII. CONCLUSION

In this paper, we tested different post sampling label distribution ratios during the data preprocessing stage on UNSW-NB15 dataset. After comparing the results of different ratios, we observed ratio13 presents the best results and applied various machine learning models on this ratio of the train set. We studied filter and embedded feature selection methods on a variety of machine learning algorithms. Our results present better performance in all models comparing to state-of-the-art research on intrusion detection. With comparable results from all machine learning models, LightGBM turns out to be the best choice for an intrusion detection system considering its high accuracy, short training time and high AUC.

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